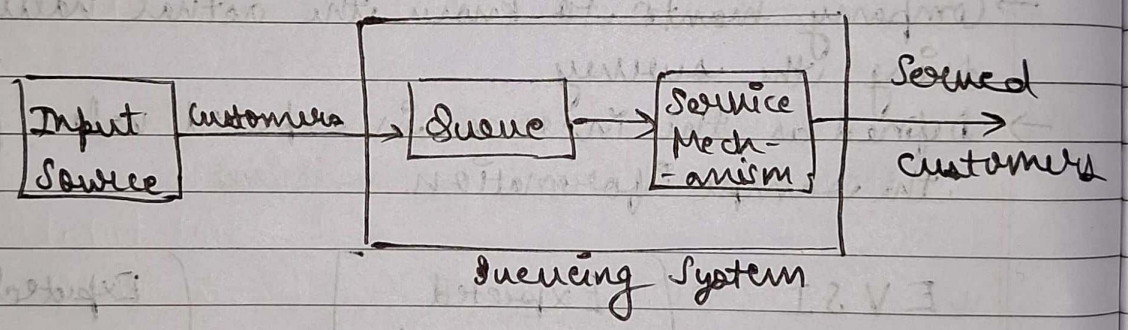
Decision Modeling
Week - 3

⇒ Queuing theory → study of waiting in many different situations.

⇒ Queuing models → determine how to operate a queue system effectively balancing cost of waiting and that of providing service.

Basic Queueing Process :



"Queue Discipline" → Rule of selecting "queue → service

Balking :- Customers refuse to join a queue seeing its length.
(walking away)

Jockeying :- Customers jumping from one queue to another.
(jumping queues)

Reneging :- Customers leave after spending some time in a queue before getting service.
(running after giving up)

"Elementary Queueing Process" → Single service utility
→ single waiting line
→ FCFS

V.V.V.

Imp. requirement of Queuing Process

$$\boxed{\text{Arrival Rate} < \text{Service Rate}}$$

Steady State performance parameters :-

P_0 = Probability of no customers in the system
P_n = " " " " " "
L = Avg. no. of customers in the system
L_q = " " " " " " queue
W = Average waiting time customers spends in system
W_q = " " " " " " queue
ρ = Utilization rate for each server
(% of time server is busy)

Little's Formula :-

① $\boxed{L = \lambda W}$ ② $\boxed{L_q = \lambda W_q}$

③ $\boxed{W = W_q + \frac{1}{\mu}}$

Kendall Notation :-

$$(a / b / c) : (d / e / f)$$

- a → Arrival dist. (Poisson)
- b → Service dist. (Exponential)
- c → No. of Servers (1/2/..∞)
- d → Service Discipline (FCFS/LIFO/SIRO)
- e → Max. no. allowed in the system (∞/finite)
- f → size of input source

Example :-

M / M / 1 / FCFS / ∞ / ∞

- M → Markovian
- M → Markovian
- ∞ → System size
- ∞ → size of input

Poisson Distribution :-

Any random phenomena that occurs on a per unit basis.

Arrival Process

- Deterministic
- Random

Poisson process

- **Orderliness**: at most 1 customer in a given time interval
- **Stationarity**: probability of arrivals within a firm interval is the same
- **Independence**: arrival of one customer has no dependence on the other

$$\boxed{P(x=n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}}$$

λ = mean arrival per time

t = length of the interval

Exponential Distribution :-

When a process is random, the time b/w the events is exponential.

$$\boxed{f(t) = \mu e^{-\mu t}}$$

μ = avg. no. of customers to be served per time period

$1/\mu$ = the mean service time

$$\boxed{P(x \le t) = 1 - e^{-\mu t}}$$

Probability that the service time X is less than time t.

⇒ 6 Properties of Exponential Distribution :-

① Strictly decreasing function :- $P(0 \le X \le \Delta t) > P(t \le X \le t + \Delta t)$

② Memoryless Property :- $\boxed{P(X > t + \Delta t \mid X > \Delta t) = P(X > t)}$

③ Min. of several independent exponential random variables has an exponential dist.

④ For all +ve values of t, $P(X \le t + \Delta t \mid X > t) = \lambda \Delta t$ for small t.

⑤ Relationship with Poisson process

If no. of arrivals is a Poisson process, then distribution of inter-arrival time is exponential.

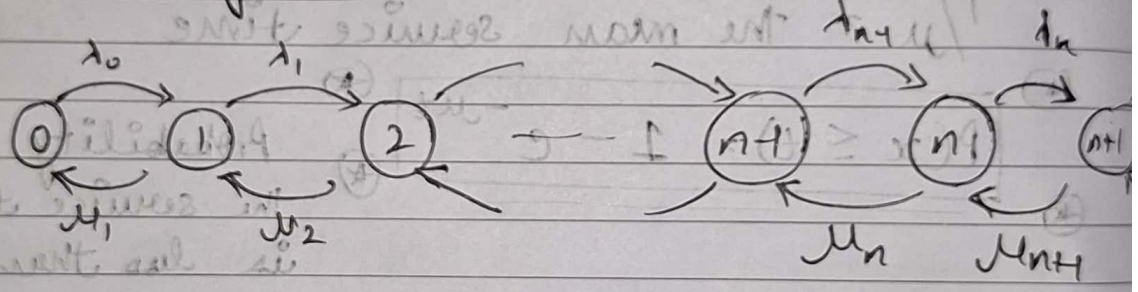
⑥ The distribution is unaffected by aggregation or disaggregation of the input process.

⊛ Birth and Death Processes :-

Birth → arrival of a customer

Death → departure of a served customer

Rate Diagram :-



(n) → state n i.e. n customers in the system

Steady State Analysis :-

① $\boxed{\text{Rate In} = \text{Rate Out}}$

② Sum of probability states = 1

$$\boxed{\sum_{i=0}^{\infty} P_i = 1}$$

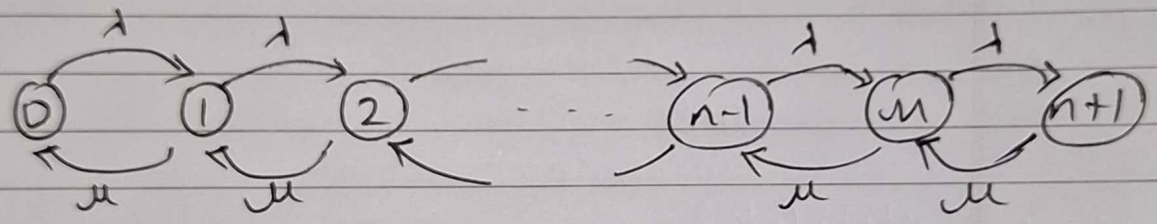
At state 0, $\lambda_0 P_0 = \mu_1 P_1$

state 1, $\lambda_0 P_0 + \mu_2 P_2 = \lambda_1 P_1 + \mu_1 P_1$

⋮

⊛ M/M/1 Model :-

$\Rightarrow$ $\lambda_n = \lambda$ and $\mu_n = \mu$ for all $n = 0, 1, 2, \ldots$



Steady state Condition :- $\boxed{\rho = \frac{\lambda}{\mu} < 1}$

$\boxed{P_0 = 1 - \rho}$ $\boxed{P_n = \rho^n (1-\rho)}$ $\boxed{P(n \geq k) = \rho^k}$

$\boxed{L = \frac{\rho}{1-\rho}}$ $\boxed{L_q = \frac{\rho^2}{1-\rho} = L - \rho}$

$\boxed{W = L/\lambda = \frac{1}{\mu - \lambda}}$ $W_q = W - 1/\mu$ from Little's law

$W = \frac{L}{\lambda}$ from Little's law $\boxed{L_q' = \frac{1}{1-\rho}}$

L_q' = Avg no. of customers in a non-empty queue